# INTRODUCTION TO DRONES

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A drone (Unmanned Aerial Vehicle) is a type of aircraft which is either autonomous or remotely piloted.

# EARLY CONCEPT

The earliest concept of UAV's was arisen by Austrians in 1894 using unmanned balloons and the first aircraft was built at the time of World War I. Technological advancements and development took place in the field of drones after the World War I.

# CATEGORIZATION OF DRONES

Drones are categorized on the basis of:

- Number of Propellers
- Size
- Range

# BASED ON NUMBER OF PROPELLERS



Tri-copters



Quad-copters



Hexa-copters



Octa-copters

# BASED ON SIZE



Nano-drones



Mini-drones



Regular-size drones



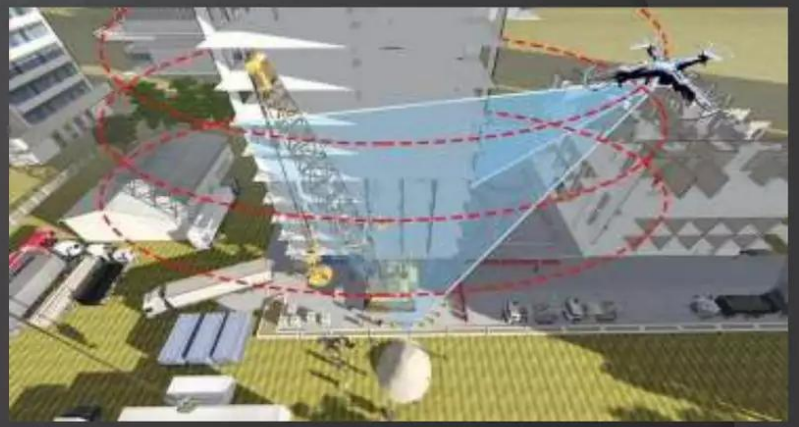
Large drones

# BASED ON RANGE

- Very close range drones
- Close range drones
- Short range drones
- Mid-range drones
- Endurance drones

# APPLICATIONS

- Construction
- Archaeology
- Precision
- Farming
- Photography





# AERODYNAMIC PRINCIPLES

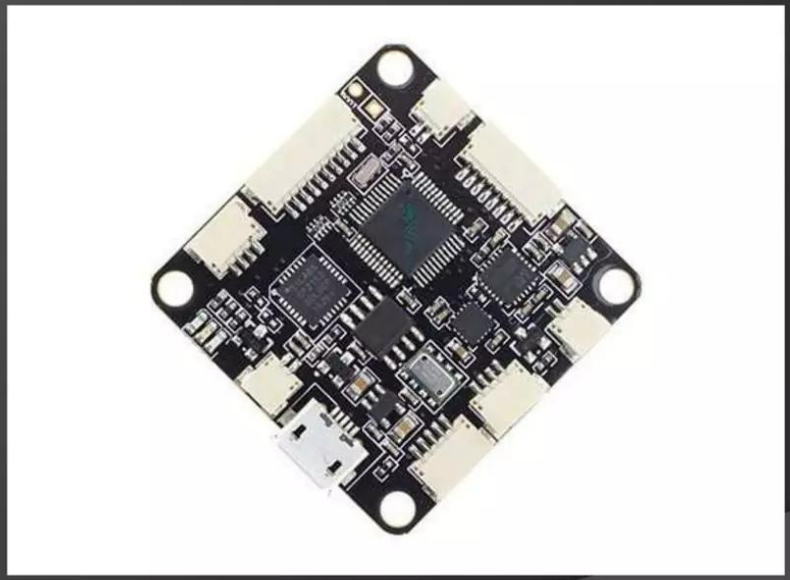
The basic aerodynamic principles may vary for different types of UAVs. The quad-copters and hexa-copters are based on the principle of Newton's laws while the aircrafts are based on the Bernoulli's principle.

# DRONE HARDWARE

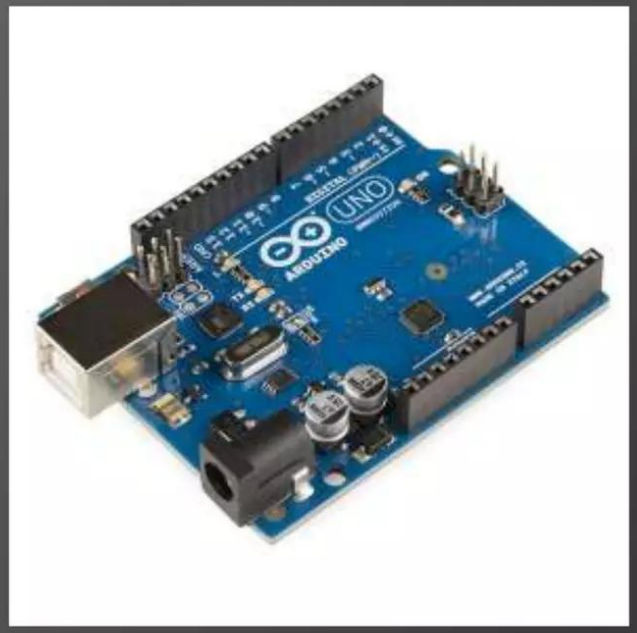
Drone's hardware includes:

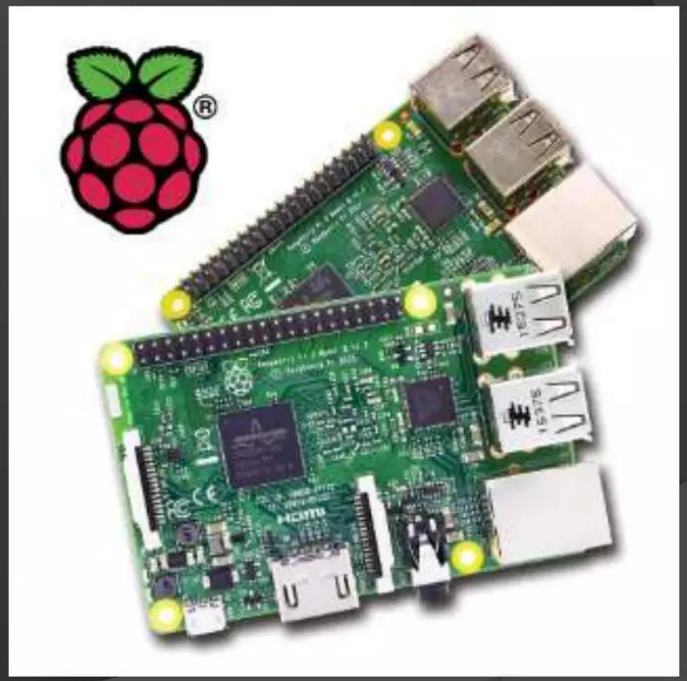
- Frame
- Propellers
- Batteries
- Transmitter-receiver pair
- Electronic Speed Controller (ESCs)
- Motors
- Flight Controller

A flight controller is pre-programmed and interfaced with certain electronic devices like gyro-meters and accelerometers to perform specific functions.



To control the hardware, electronic modules like Arduino, Raspberry Pi could also be used for self-programming.



The type of propeller (the fan that transmits power by converting rotational motion into thrust) depends on the function that the drone has to perform. These propellers differentiate on the basis of shape and size.





The size of the frame completely depends on the needs. For example, for aerial cinematography, a big frame is used to lift camera with tall landing gears and for high maneuverability, small frame is the best option.



The motors, battery and ESC's to be chosen depend on the weight that drone needs to carry.






The appropriate transmitters and receivers are required. For example, minimum 4-channel transmitter and receiver for a quad-copter.






# DRONE CONTROLS

The drone controls involve the mathematical functions of PID controls (Proportional-Integral-Derivative control) and the Thrust-to-Weight ratio. The position and orientation of the robot are defined by the terms rolling, pitching and yawing (steering).

# ADVANCED DRONES AVAILABLE TODAY

- DJI Inspire 1 Pro Quad-copter
- ALTA UAV Drone
- Faucon's Drone
- Passport (Hover Camera), etc.

# SCOPE OF DRONES

The functions of drones are becoming more advanced with time, the scope of UAV's is widening and it shall have applications in our day-to-day lives in near future.




Drone




Parrot



# THANK YOU